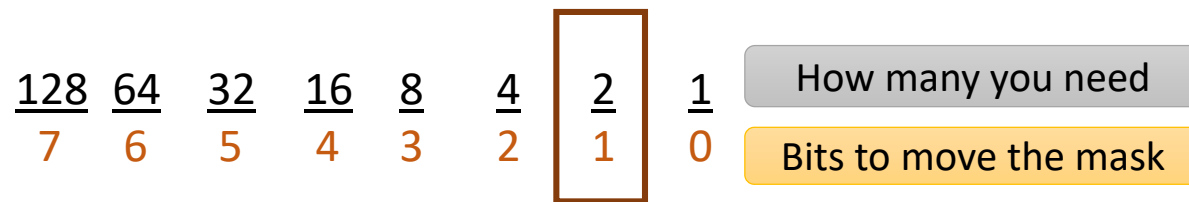


Moving the Subnet Mask

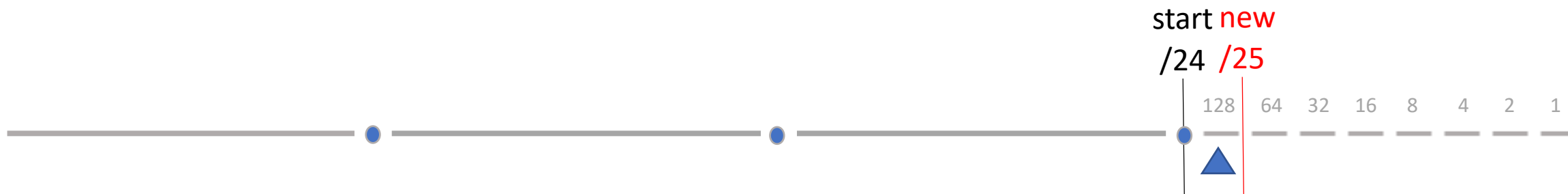


Starting mask /24
255.255.255.0

Need 2 subnets
Move mask 1 bit

New mask /25
255.255.255.128

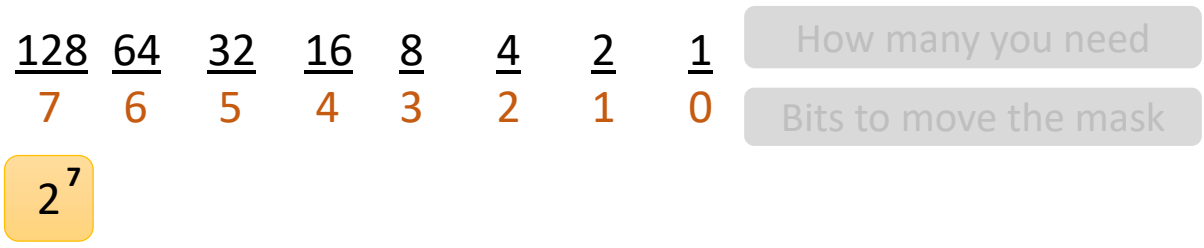
Delta = 128
In the 4th Octet
Subnet IDs increment by 128



1. Figure out the starting subnet mask
2. Use the subnet calculator to figure out the new subnet mask
3. Figure out the delta, and the octet that the delta is in
4. Calculate subnet IDs based on the delta

1 bit for
subnets
 $2^1 = 2$

7 bits for
hosts
 $2^7 - 2 = 126$

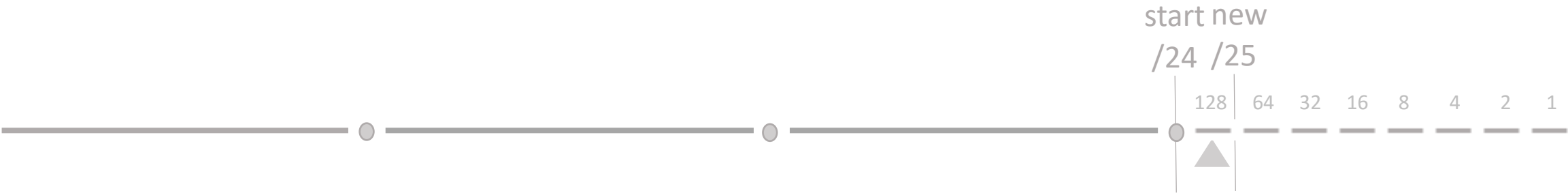


Starting mask /24
255.255.255.0

Need 2 subnets
Move mask 1 bit

New mask /25
255.255.255.128

Delta = 128
In the 4th Octet
Subnet IDs increment by 128



- 1. Figure out the starting subnet mask
- 2. Use the subnet calculator to figure out the new subnet mask
- 3. Figure out the delta, and the octet that the delta is in
- 4. Calculate subnet IDs based on the delta

1 bit for subnets
2¹ = 2

3 bits for hosts
2³ - 2 = 6

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

How many you need

Bits to move the mask

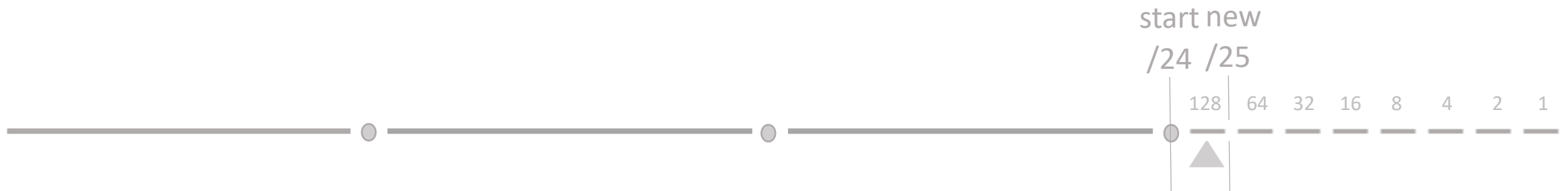
2^6

Starting mask /24
255.255.255.0

Need 2 subnets
Move mask 1 bit

New mask /25
255.255.255.128

Delta = 128
In the 4th Octet
Subnet IDs increment by 128



1. Figure out the starting subnet mask
2. Use the subnet calculator to figure out the new subnet mask
3. Figure out the delta, and the octet that the delta is in
4. Calculate subnet IDs based on the delta

1 bit for
subnets
 $2^1 = 2$

3 bits for
hosts
 $2^3 - 2 = 6$

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

How many you need

Bits to move the mask

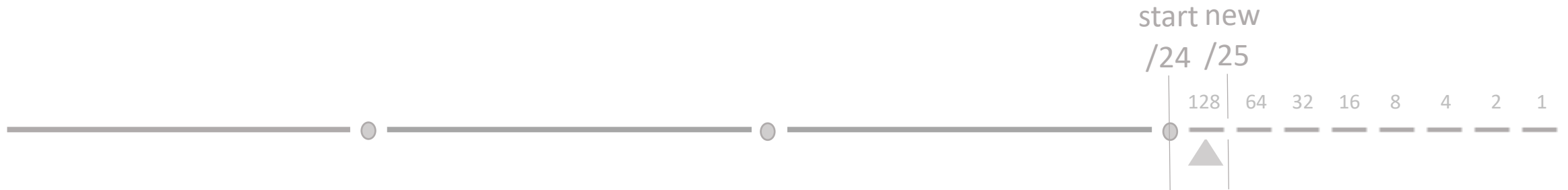
2^5

Starting mask /24
255.255.255.0

Need 2 subnets
Move mask 1 bit

New mask /25
255.255.255.128

Delta = 128
In the 4th Octet
Subnet IDs increment by 128



1. Figure out the starting subnet mask
2. Use the subnet calculator to figure out the new subnet mask
3. Figure out the delta, and the octet that the delta is in
4. Calculate subnet IDs based on the delta

1 bit for
subnets
 $2^1 = 2$

3 bits for
hosts
 $2^3 - 2 = 6$

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
7	6	5	4	3	2	1	0

How many you need

Bits to move the mask

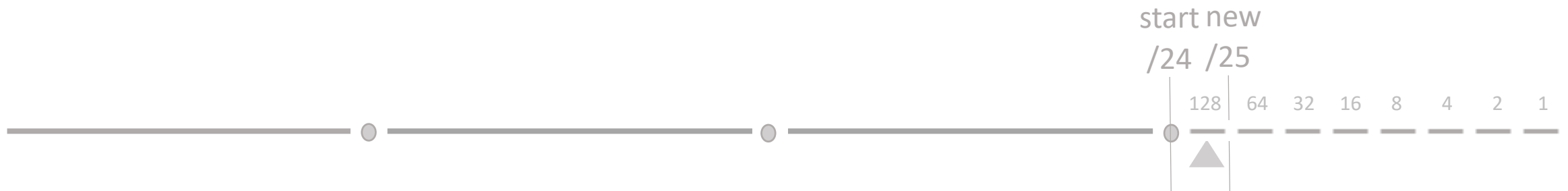
2^4

Starting mask /24
255.255.255.0

Need 2 subnets
Move mask 1 bit

New mask /25
255.255.255.128

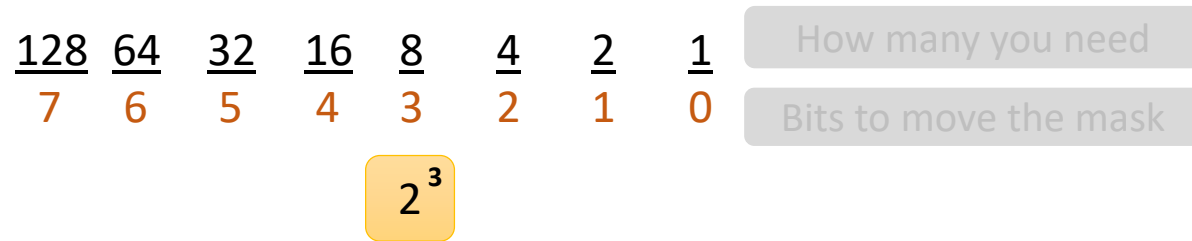
Delta = 128
In the 4th Octet
Subnet IDs increment by 128



1. Figure out the starting subnet mask
2. Use the subnet calculator to figure out the new subnet mask
3. Figure out the delta, and the octet that the delta is in
4. Calculate subnet IDs based on the delta

1 bit for
subnets
 $2^1 = 2$

3 bits for
hosts
 $2^3 - 2 = 6$

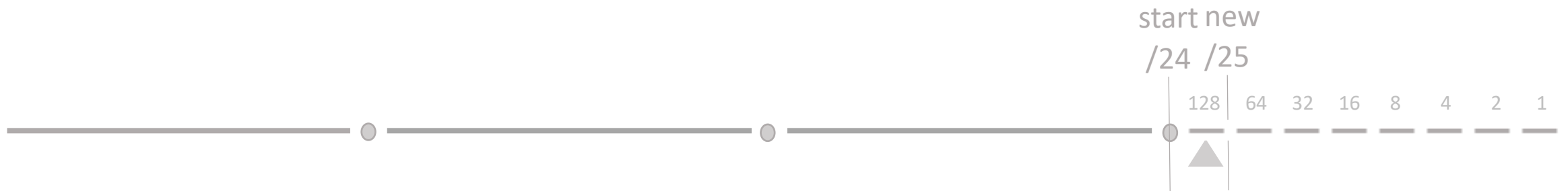


Starting mask /24
255.255.255.0

Need 2 subnets
Move mask 1 bit

New mask /25
255.255.255.128

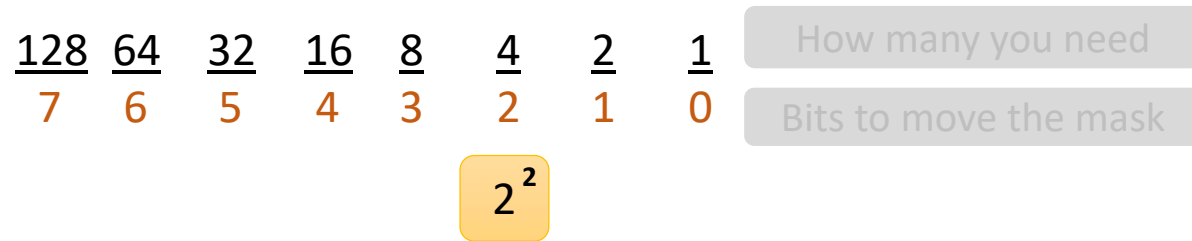
Delta = 128
In the 4th Octet
Subnet IDs increment by 128



1. Figure out the starting subnet mask
2. Use the subnet calculator to figure out the new subnet mask
3. Figure out the delta, and the octet that the delta is in
4. Calculate subnet IDs based on the delta

1 bit for
subnets
 $2^1 = 2$

3 bits for
hosts
 $2^3 - 2 = 6$

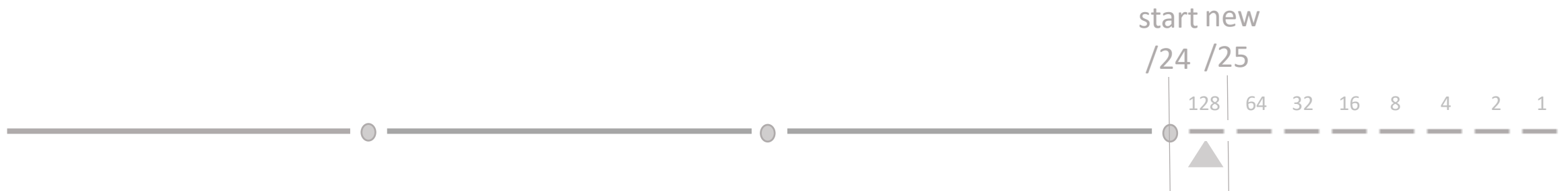


Starting mask /24
255.255.255.0

Need 2 subnets
Move mask 1 bit

New mask /25
255.255.255.128

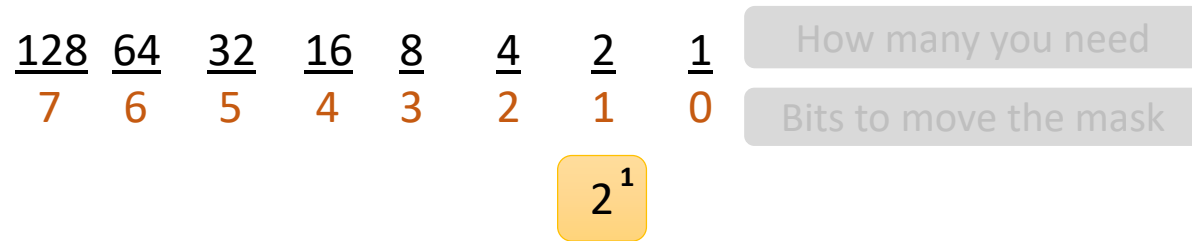
Delta = 128
In the 4th Octet
Subnet IDs increment by 128



1. Figure out the starting subnet mask
2. Use the subnet calculator to figure out the new subnet mask
3. Figure out the delta, and the octet that the delta is in
4. Calculate subnet IDs based on the delta

1 bit for
subnets
 $2^1 = 2$

3 bits for
hosts
 $2^3 - 2 = 6$

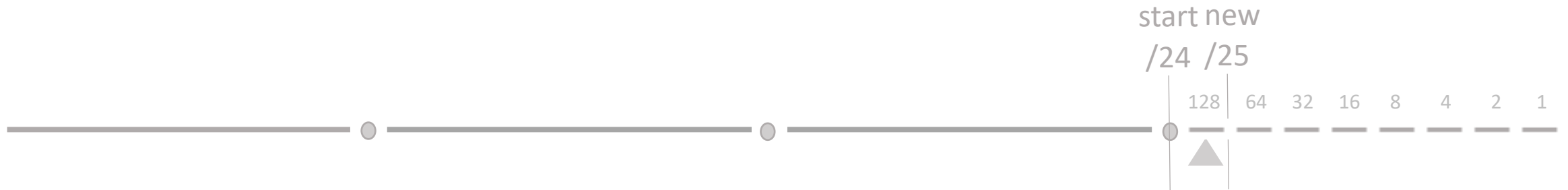


Starting mask /24
255.255.255.0

Need 2 subnets
Move mask 1 bit

New mask /25
255.255.255.128

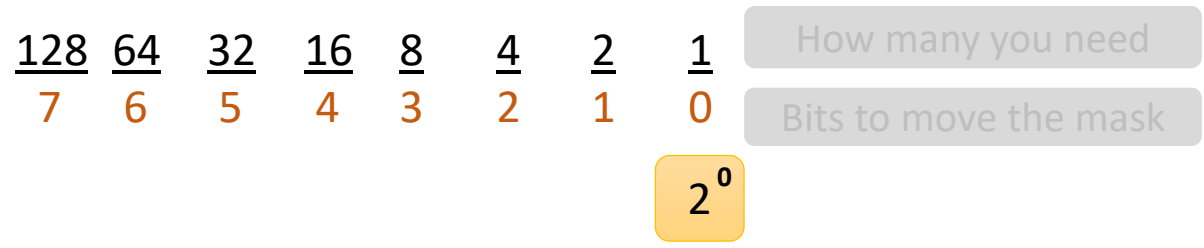
Delta = 128
In the 4th Octet
Subnet IDs increment by 128



1. Figure out the starting subnet mask
2. Use the subnet calculator to figure out the new subnet mask
3. Figure out the delta, and the octet that the delta is in
4. Calculate subnet IDs based on the delta

1 bit for
subnets
 $2^1 = 2$

3 bits for
hosts
 $2^3 - 2 = 6$



Starting mask /24
255.255.255.0

Need 2 subnets
Move mask 1 bit

New mask /25
255.255.255.128

Delta = 128
In the 4th Octet
Subnet IDs increment by 128

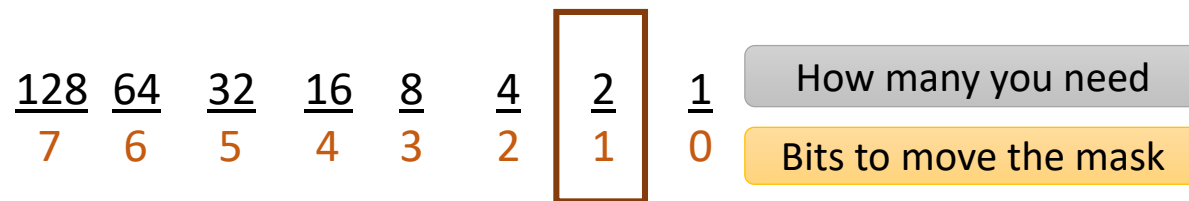


1. Figure out the starting subnet mask
2. Use the subnet calculator to figure out the new subnet mask
3. Figure out the delta, and the octet that the delta is in
4. Calculate subnet IDs based on the delta

1 bit for
subnets
 $2^1 = 2$

3 bits for
hosts
 $2^3 - 2 = 6$

Going back to where we were...

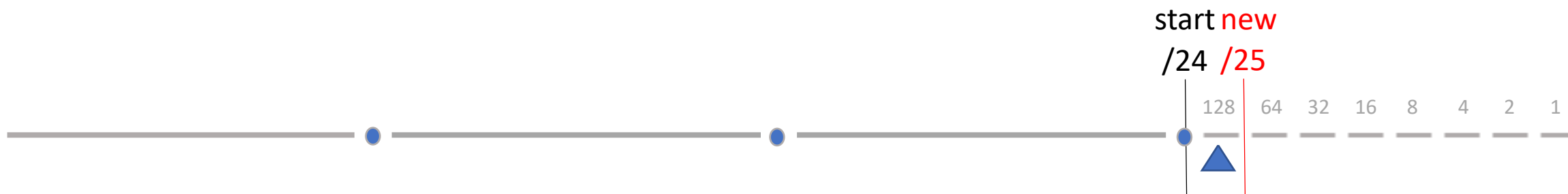


Starting mask /24
255.255.255.0

Need 2 subnets
Move mask 1 bit

New mask /25
255.255.255.128

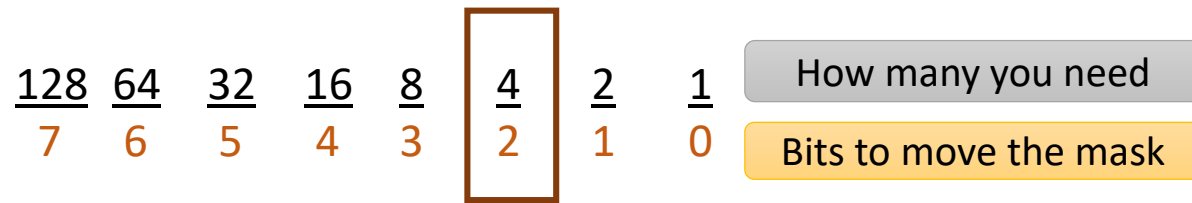
Delta = 128
In the 4th Octet
Subnet IDs increment by 128



1. Figure out the starting subnet mask
2. Use the subnet calculator to figure out the new subnet mask
3. Figure out the delta, and the octet that the delta is in
4. Calculate subnet IDs based on the delta

1 bit for
subnets
 $2^1 = 2$

7 bits for
hosts
 $2^7 - 2 = 126$

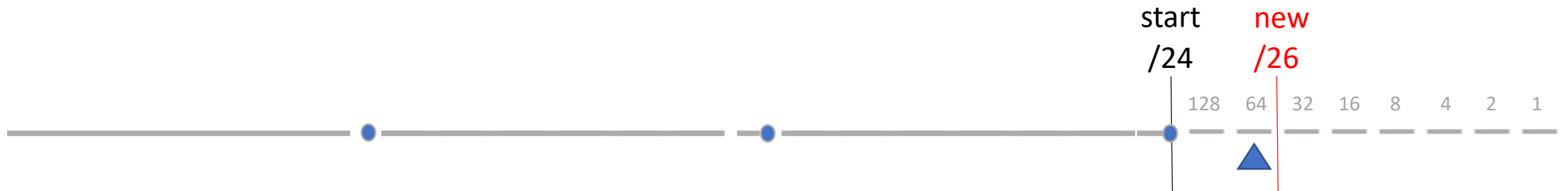


Starting mask /24
255.255.255.0

Need 4 subnets
Move mask 2 bits

New mask /26
255.255.255.192

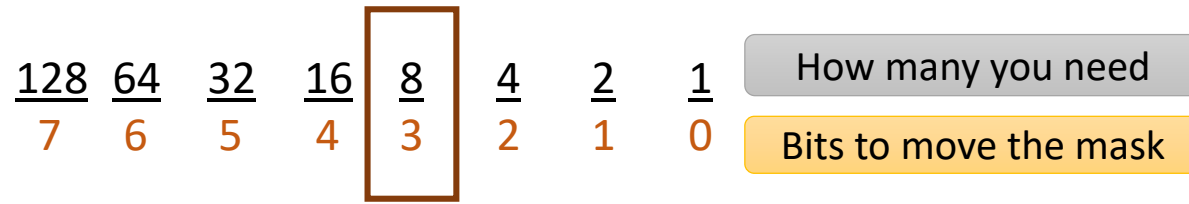
Delta = 64
In the 4th Octet
Subnet IDs increment by 64



1. Figure out the starting subnet mask
2. Use the subnet calculator to figure out the new subnet mask
3. Figure out the delta, and the octet that the delta is in
4. Calculate subnet IDs based on the delta

2 bits for
subnets
 $2^2 = 4$

6 bits for
hosts
 $2^6 - 2 = 62$

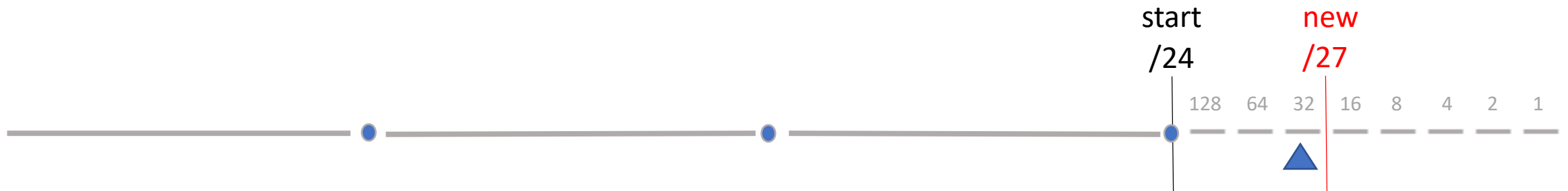


Starting mask /24
255.255.255.0

Need 8 subnets
Move mask 3 bits

New mask /27
255.255.255.224

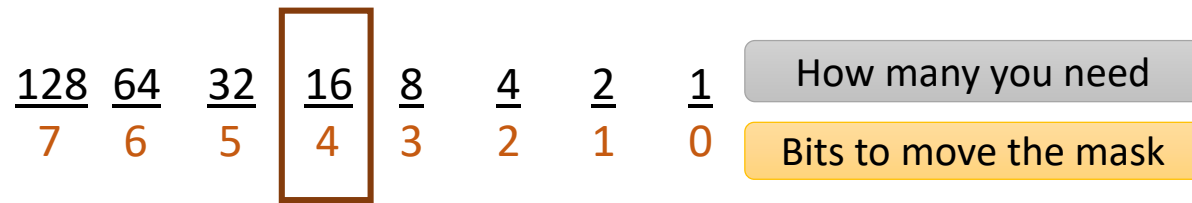
Delta = 32
In the 4th Octet
Subnet IDs increment by 32



1. Figure out the starting subnet mask
2. Use the subnet calculator to figure out the new subnet mask
3. Figure out the delta, and the octet that the delta is in
4. Calculate subnet IDs based on the delta

3 bits for
subnets
 $2^3 = 8$

5 bits for
hosts
 $2^5 - 2 = 30$

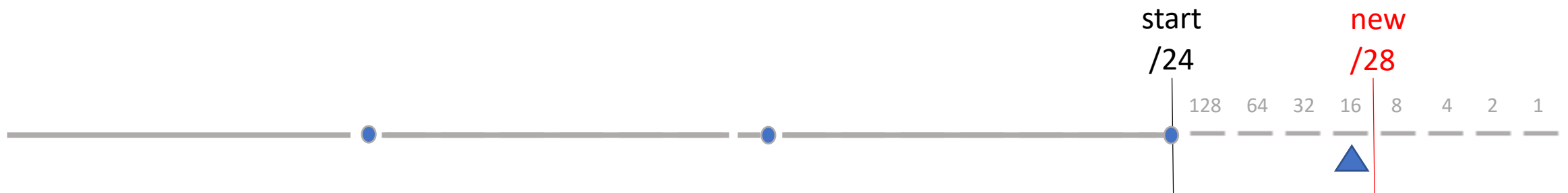


Starting mask /24
255.255.255.0

Need 16 subnets
Move mask 4 bits

New mask /28
255.255.255.240

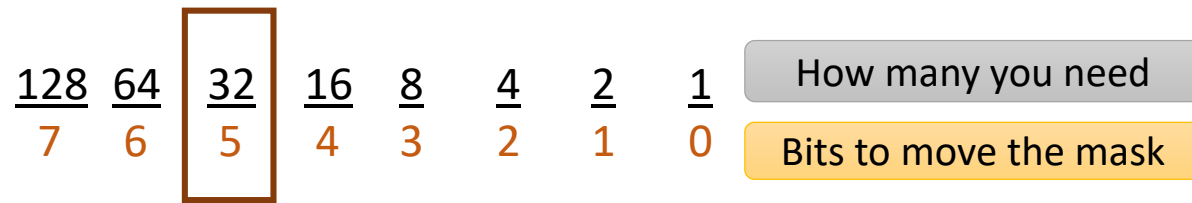
Delta = 16
In the 4th Octet
Subnet IDs increment by 16



1. Figure out the starting subnet mask
2. Use the subnet calculator to figure out the new subnet mask
3. Figure out the delta, and the octet that the delta is in
4. Calculate subnet IDs based on the delta

4 bits for
subnets
 $2^4 = 16$

4 bits for
hosts
 $2^4 - 2 = 14$

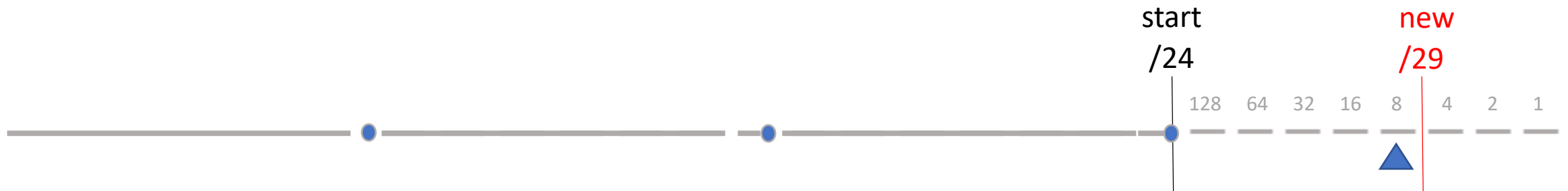


Starting mask /24
255.255.255.0

Need 32 subnets
Move mask 5 bits

New mask /29
255.255.255.248

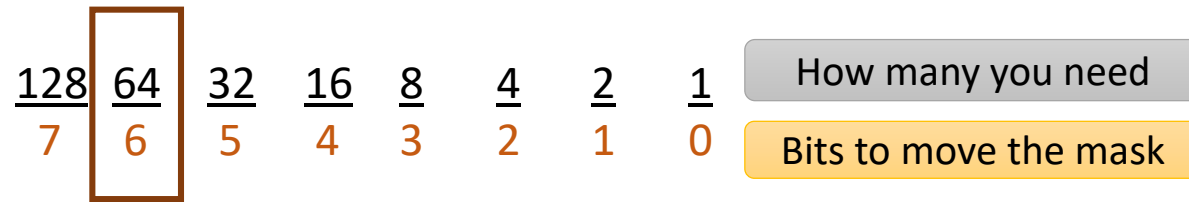
Delta = 8
In the 4th Octet
Subnet IDs increment by 8



1. Figure out the starting subnet mask
2. Use the subnet calculator to figure out the new subnet mask
3. Figure out the delta, and the octet that the delta is in
4. Calculate subnet IDs based on the delta

5 bits for
subnets
 $2^5 = 32$

3 bits for
hosts
 $2^3 - 2 = 6$

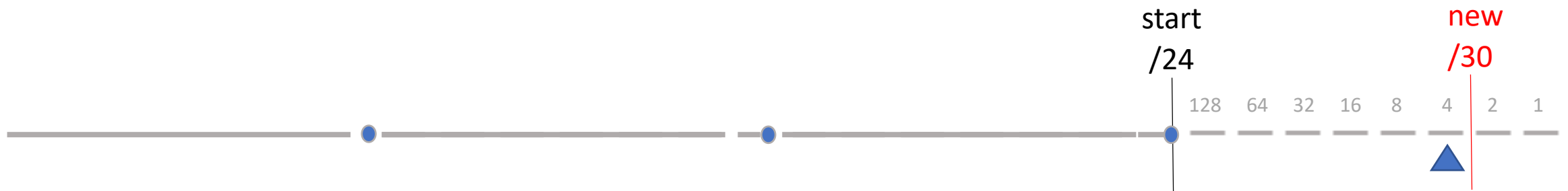


Starting mask /24
255.255.255.0

Need 64 subnets
Move mask 6 bits

New mask /30
255.255.255.252

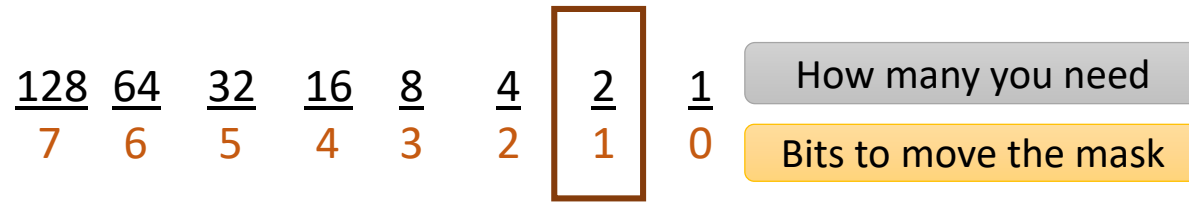
Delta = 4
In the 4th Octet
Subnet IDs increment by 4



1. Figure out the starting subnet mask
2. Use the subnet calculator to figure out the new subnet mask
3. Figure out the delta, and the octet that the delta is in
4. Calculate subnet IDs based on the delta

6 bits for
subnets
 $2^6 = 64$

2 bits for
hosts
 $2^2 - 2 = 2$

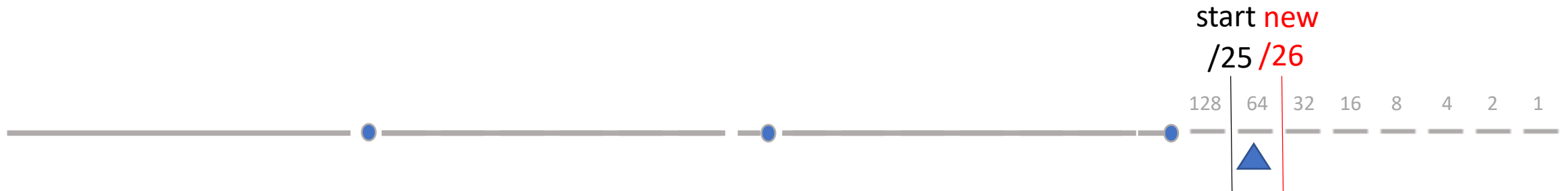


Starting mask /25
255.255.255.128

Need 2 subnets
Move mask 1 bit

New mask /26
255.255.255.192

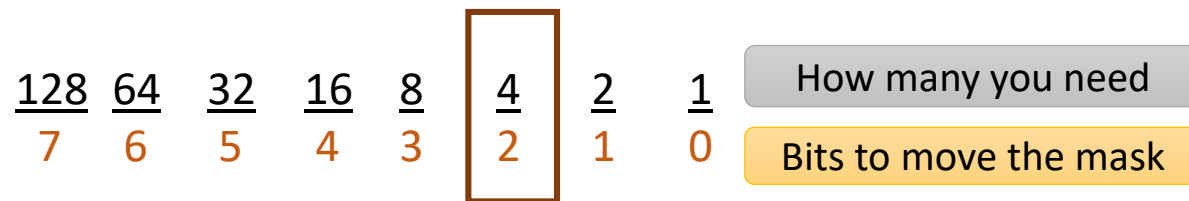
Delta = 64
In the 4th Octet
Subnet IDs increment by 64



1. Figure out the starting subnet mask
2. Use the subnet calculator to figure out the new subnet mask
3. Figure out the delta, and the octet that the delta is in
4. Calculate subnet IDs based on the delta

1 bit for
subnets
 $2^1 = 2$

6 bits for
hosts
 $2^6 - 2 = 62$

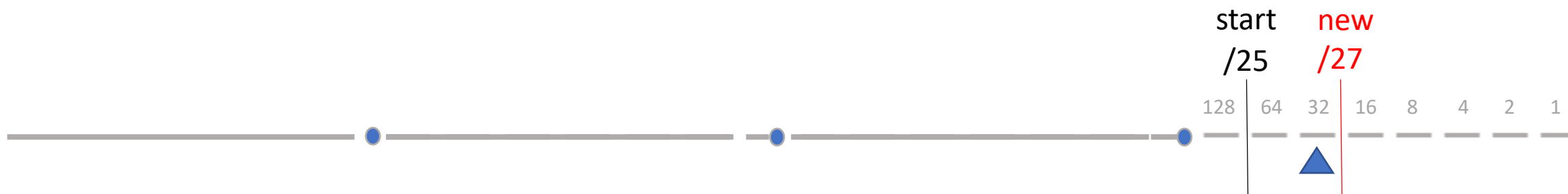


Starting mask /25
255.255.255.128

Need 4 subnets
Move mask 2 bits

New mask /27
255.255.255.224

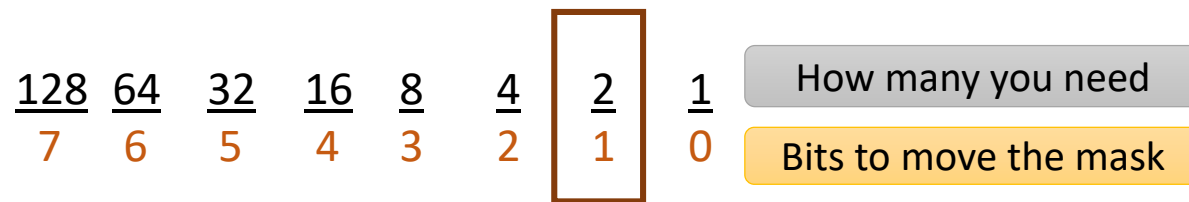
Delta = 32
In the 4th Octet
Subnet IDs increment by 32



1. Figure out the starting subnet mask
2. Use the subnet calculator to figure out the new subnet mask
3. Figure out the delta, and the octet that the delta is in
4. Calculate subnet IDs based on the delta

2 bits for
subnets
 $2^2 = 4$

5 bits for
hosts
 $2^5 - 2 = 30$

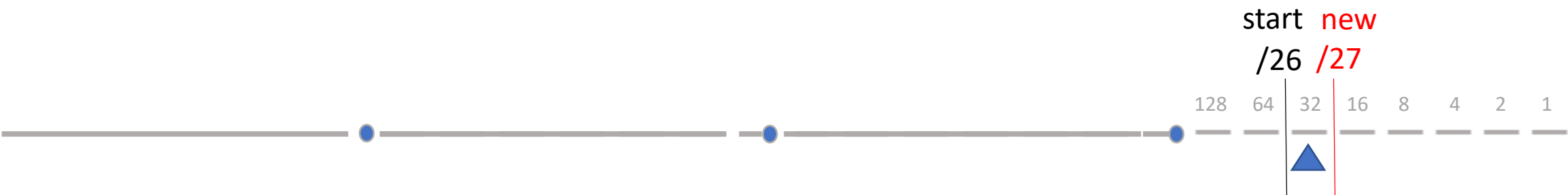


Starting mask /26
255.255.255.192

Need 2 subnets
Move mask 1 bit

New mask /27
255.255.255.224

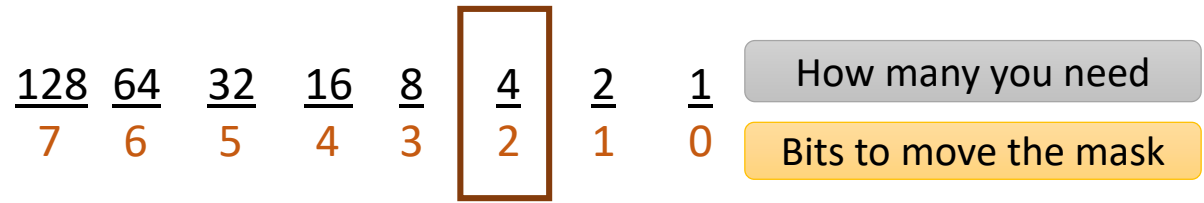
Delta = 32
In the 4th Octet
Subnet IDs increment by 32



1. Figure out the starting subnet mask
2. Use the subnet calculator to figure out the new subnet mask
3. Figure out the delta, and the octet that the delta is in
4. Calculate subnet IDs based on the delta

1 bit for subnets
 $2^1 = 2$

5 bits for hosts
 $2^5 - 2 = 30$

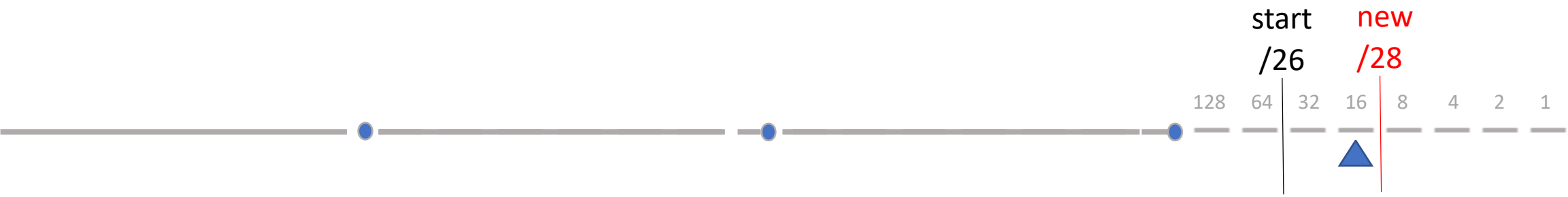


Starting mask /26
255.255.255.192

Need 4 subnets
Move mask 2 bits

New mask /28
255.255.255.240

Delta = 16
In the 4th Octet
Subnet IDs increment by 16



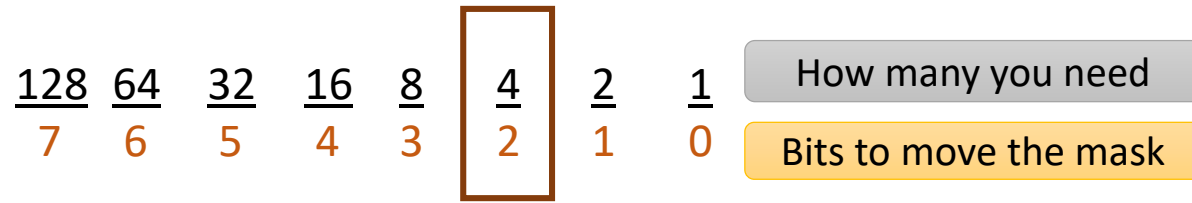
1. Figure out the starting subnet mask
2. Use the subnet calculator to figure out the new subnet mask
3. Figure out the delta, and the octet that the delta is in
4. Calculate subnet IDs based on the delta

2 bits for subnets

$2^2 = 4$

4 bits for hosts

$2^4 - 2 = 14$

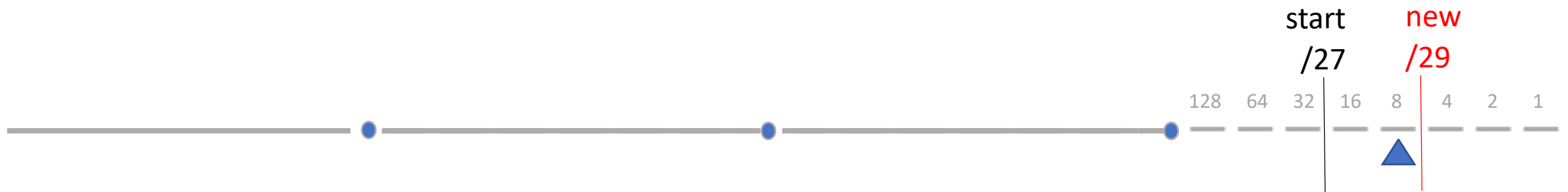


Starting mask /27
255.255.255.224

Need 4 subnets
Move mask 2 bits

New mask /29
255.255.255.248

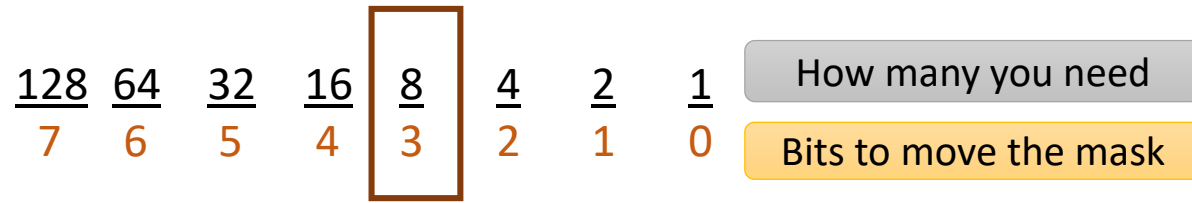
Delta = 8
In the 4th Octet
Subnet IDs increment by 8



1. Figure out the starting subnet mask
2. Use the subnet calculator to figure out the new subnet mask
3. Figure out the delta, and the octet that the delta is in
4. Calculate subnet IDs based on the delta

2 bits for
subnets
 $2^2 = 4$

3 bits for
hosts
 $2^3 - 2 = 6$

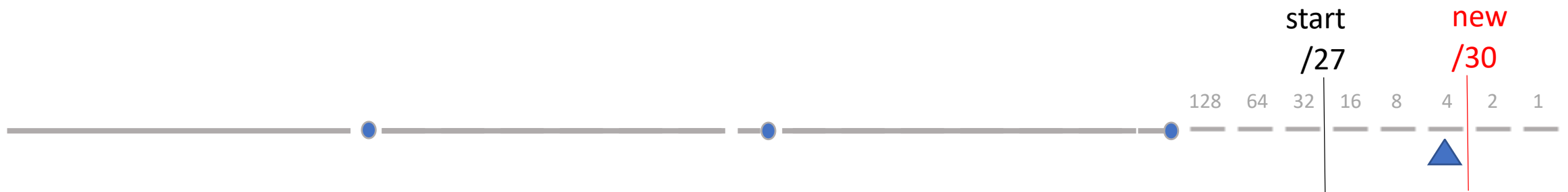


Starting mask /27
255.255.255.224

Need 8 subnets
Move mask 3 bits

New mask /30
255.255.255.252

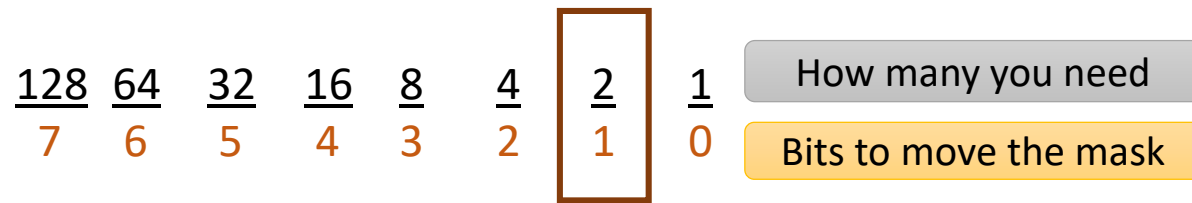
Delta = 4
In the 4th Octet
Subnet IDs increment by 4



1. Figure out the starting subnet mask
2. Use the subnet calculator to figure out the new subnet mask
3. Figure out the delta, and the octet that the delta is in
4. Calculate subnet IDs based on the delta

3 bits for
subnets
 $2^3 = 8$

2 bits for
hosts
 $2^2 - 2 = 2$

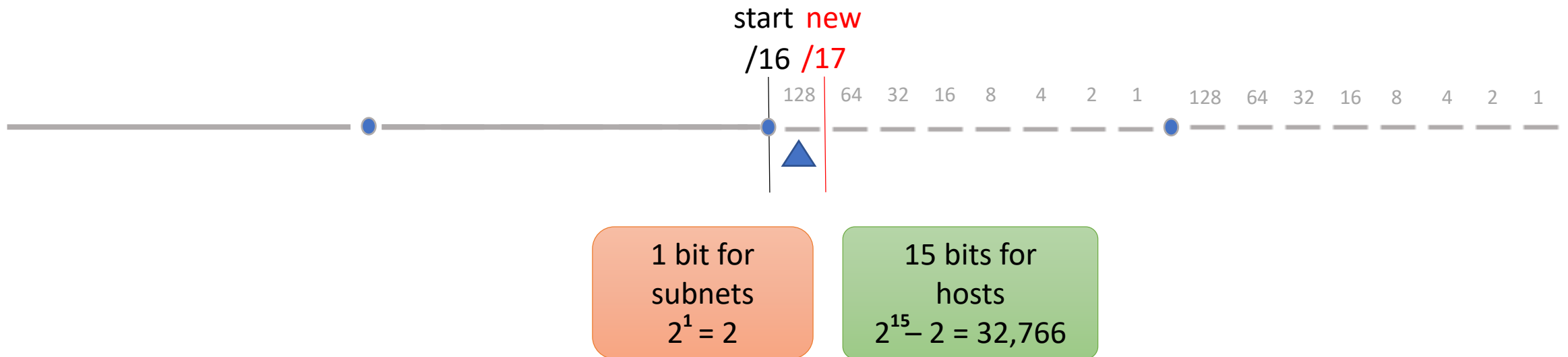


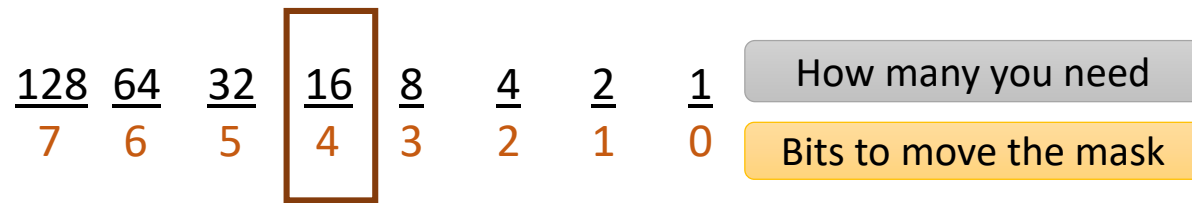
Starting mask /16
255.255.0.0

Need 2 subnets
Move mask 1 bits

New mask /17
255.255.128.0

Delta = 128
In the 3rd Octet
Subnet IDs increment by 128



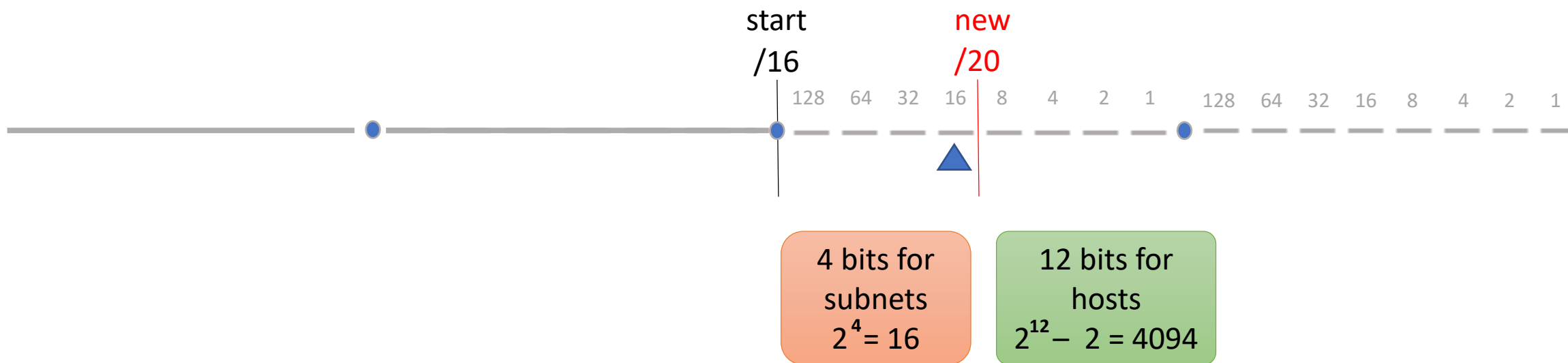


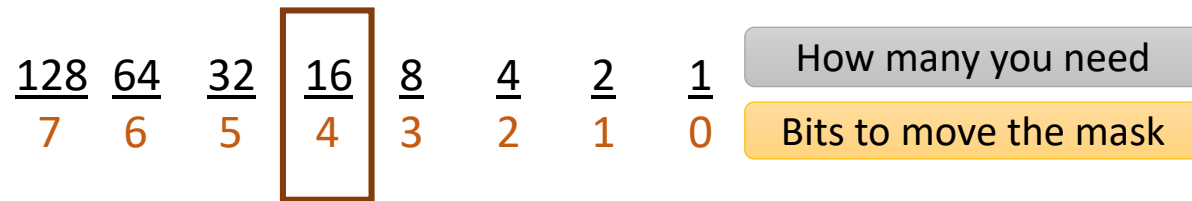
Starting mask /16
255.255.0.0

Need 16 subnets
Move mask 4 bits

New mask /20
255.255.240.0

Delta = 16
In the 3rd Octet
Subnet IDs increment by 16



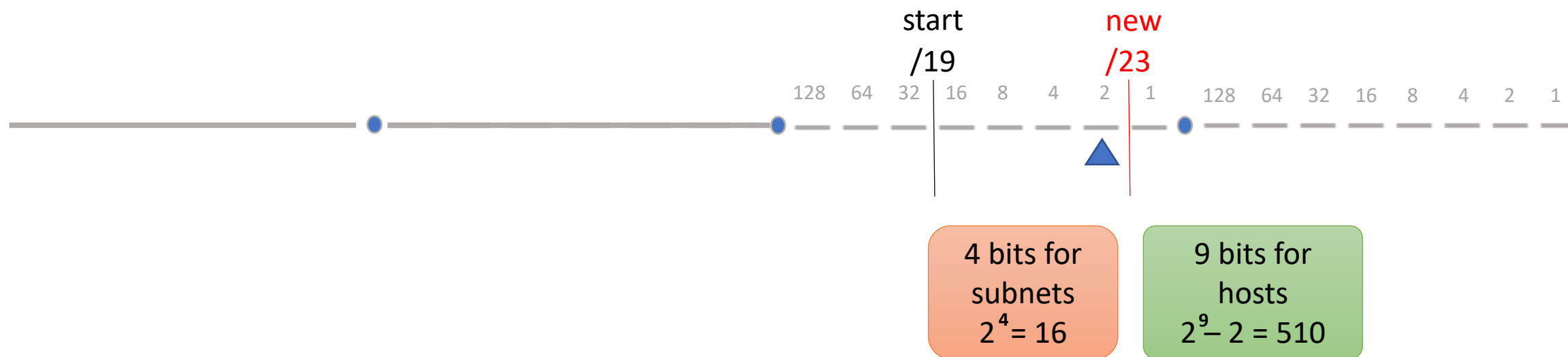


Starting mask /19
255.255.224.0

Need 16 subnets
Move mask 4 bit

New mask /23
255.255.254.0

Delta = 2
In the 3rd Octet
Subnet IDs increment by 2



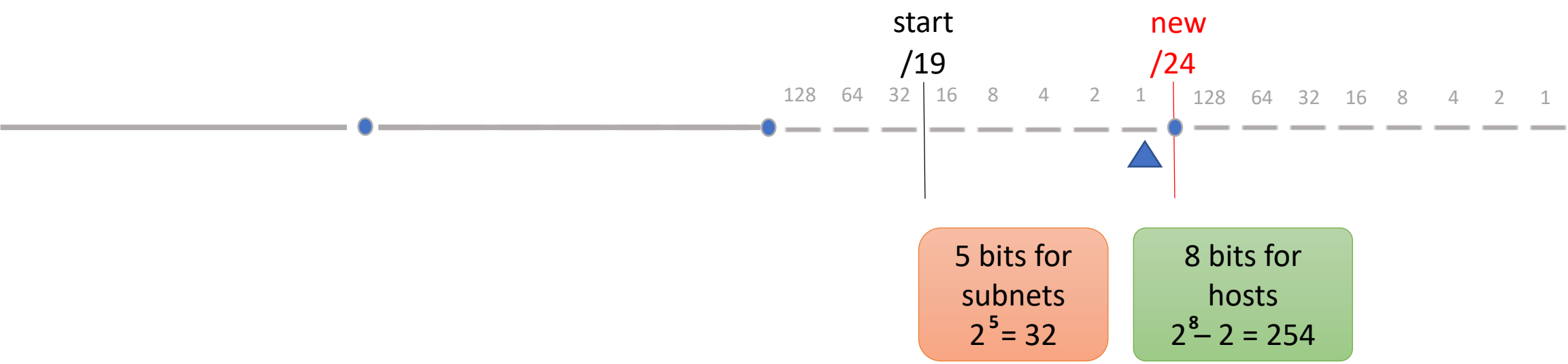
<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>	How many you need
7	6	5	4	3	2	1	0	

Starting mask /19
255.255.224.0

Need 32 subnets
Move mask 5 bit

New mask /24
255.255.255.0

Delta = 1
In the 3rd Octet
Subnet IDs increment by 1



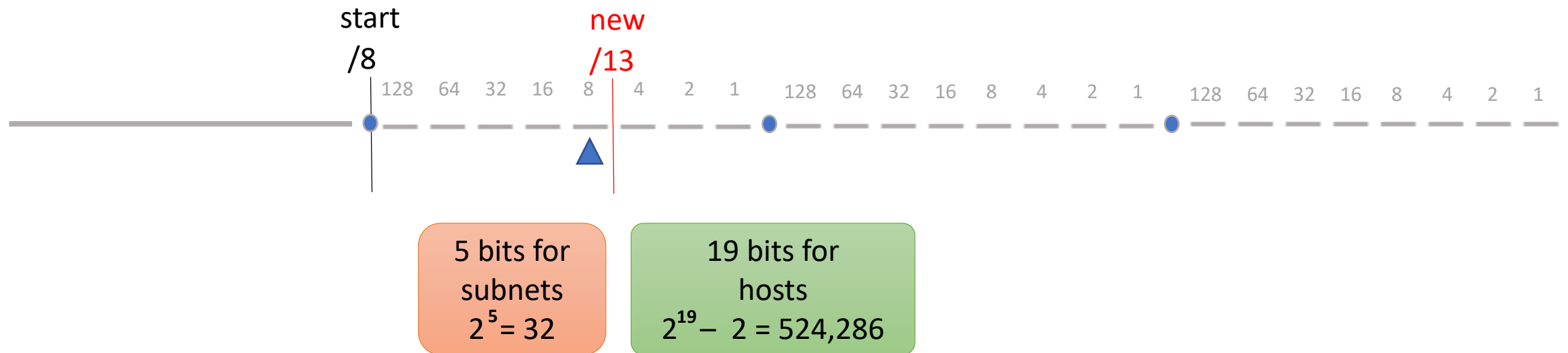
<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>	How many you need
7	6	5	4	3	2	1	0	Bits to move the mask

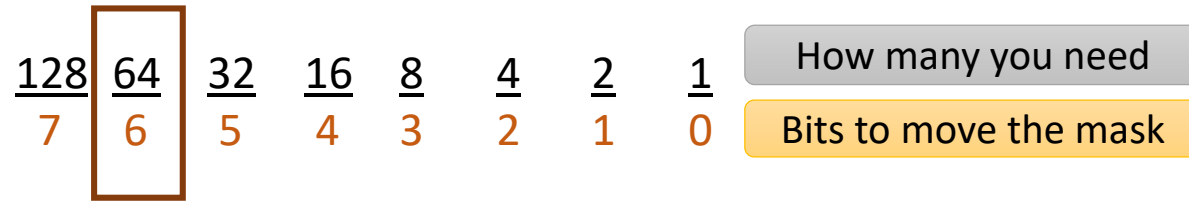
Starting mask /8
255.0.0.0

Need 32 subnets
Move mask 5 bit

New mask /13
255.248.0.0

Delta = 8
In the 2nd octet
Subnet IDs increment by 8



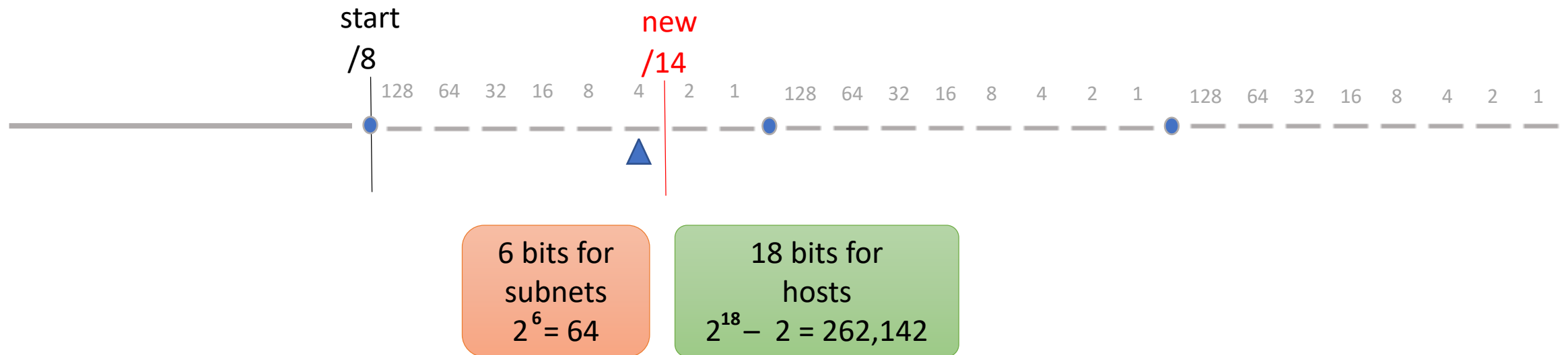


Starting mask /8
255.0.0.0

Need 64 subnets
Move mask 6 bit

New mask /14
255.252.0.0

Delta = 4
In the 2nd octet
Subnet IDs increment by 4



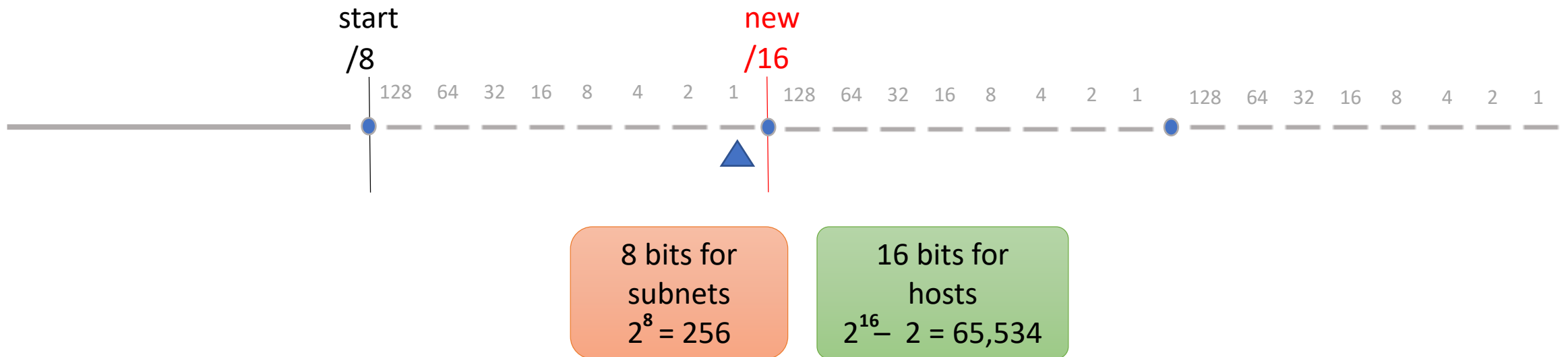
32,768	16,384	8192	4096	2048	1024	512	256	128	64	32	16	8	4	2	1	How many you need
<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>	<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Bits to move the mask

Starting mask /8
255.0.0.0

Need 250 subnets
Move mask 8 bits

New mask /16
255.255.0.0

Delta = 1
In the 2nd octet
Subnet IDs increment by 1



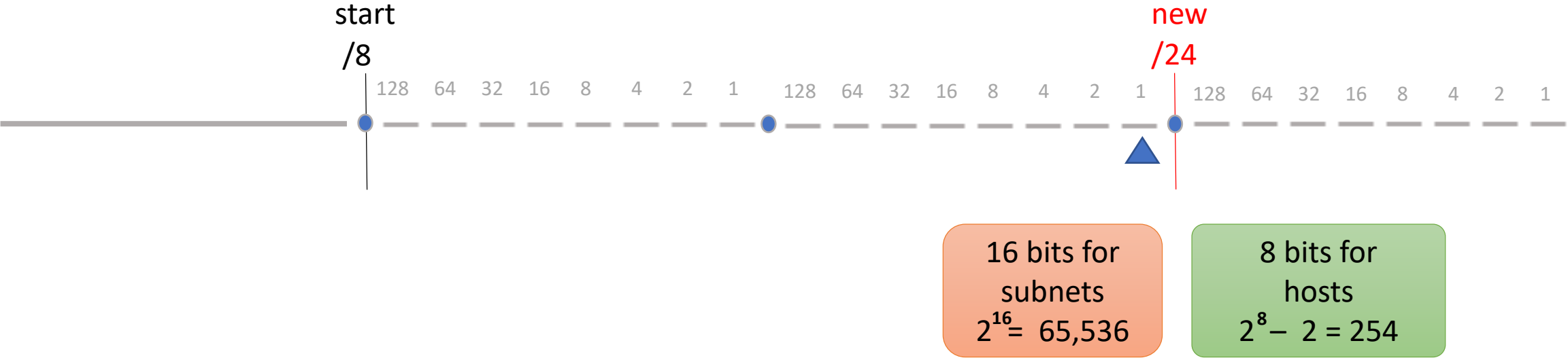
65,536	32,768	16,384	8192	4096	2048	1024	512	256	128	64	32	16	8	4	2	1	How many you need
<u>1</u>	<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>	<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>	Bits to move the mask
16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	

Starting mask /8
255.0.0.0

Need 50,000 subnets
Move mask 16 bits

New mask /24
255.255.255.0

Delta = 1
In the 3rd octet
Subnet IDs increment by 1



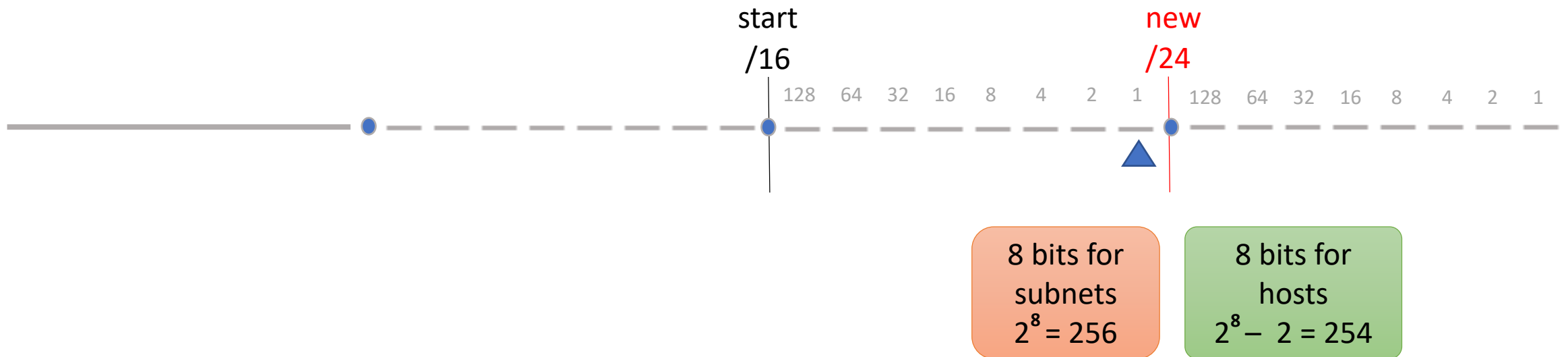
32,768	16,384	8192	4096	2048	1024	512	256	128	64	32	16	8	4	2	1	How many you need
<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>	<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>	Bits to move the mask
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	

Starting mask /16
255.255.0.0

Need 250 subnets
Move mask 8 bits

New mask /24
255.255.255.0

Delta = 1
In the 3rd octet
Subnet IDs increment by 1



32,768	16,384	8192	4096	2048	1024	512	256	128	64	32	16	8	4	2	1	How many you need
<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>	<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>	Bits to move the mask
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	

Starting mask /12
255.240.0.0

Need 250 subnets
Move mask 8 bits

New mask /20
255.255.240.0

Delta = 16
In the 3rd octet
Subnet IDs increment by 16

